Theory of the Beginning in the PSP Model

E. V. Chernoknizhny

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Abstract

This work presents a complete theory of the beginning of the cycle in the PSP (Phase State Parameter) model. The transition from quantum vacuum fluctuation to the birth of the torus-flywheel, the outer shell, and the launch of cyclic motion is described. The theory includes: quantum-field justification of the origin of protomatter, the interaction Lagrangian, the sequence of phase transitions from $M = -0.8$ to $M = 0$, the origin of the global rhythm $T_0 = 16.35$ days, the connection with general relativity, comparison with ΛCDM, and testable predictions.

The numerical values of 16.35 days and 0.60 days are used solely for readability and reader comprehension. In the strict formulation of the PSP model, all quantities are dimensionless and expressed through the phase M, the phase flow Φ_M, and the shell radius R_{shell}.

1 Introduction

The standard cosmological model ΛCDM successfully describes a wide range of observational data but leaves fundamental questions unanswered: what was before the Big Bang, why is the Universe so homogeneous, what are dark matter and dark energy? This work proposes an alternative model — PSP (Phase State Parameter) — in which time is replaced by the cycle phase M. The Universe is described as a three-level closed structure: an outer shell (dark energy), a cavity (matter and fields), and an inner torus-flywheel with a quasar.

2 Definitions and Basic Assumptions

Table 1: Basic definitions of the PSP model.

Quantity	Definition
M	Dimensionless cycle phase. $M \in [-0.8, 1]$. $M = 0$ is the assembly point of the cycle.
ψ	Field of charged units of protomatter.
Φ	Scalar gravitational potential.
A_μ	Vector potential of the electromagnetic field.
Φ_M	Phase flow: $\Phi_M = dM/dz$.
R_{shell}	Radius of the outer shell ($\approx$ 26.1 billion light-years, in projection).

Basic assumptions:

1. Protomatter is in an inert state before the first impulse.
2. Fields (gravitational, magnetic, electronic) are inert before oscillations.
3. All stages obey the laws of conservation and thermodynamics.
4. In the local limit $M \to 0$, the PSP metric reduces to the Minkowski metric.
5. The model does not require dark matter or dark energy as separate entities.

3 The Meaning of Negative M — State of Rest

In the PSP model, negative values of M ($M < 0$) correspond to a state of rest (the absence of cyclic motion). In this state, the system has no phase as a coordinate; its evolution is described exclusively by field dynamics.

Table 2: Meaning of M at different stages.

Range of M	Meaning
$M < 0$	State of rest. No cycle.
$M = -0.8$	Beginning of beginnings. First fluctuation.
$M = 0$	Assembly point. Birth of the cycle.
$M > 0$	Cyclic motion.

4 Origin of Protomatter and the Field ψ

The field ψ describes the collective state of charged units of protomatter arising from a quantum vacuum fluctuation at the stage $M = -0.8$.

In quantum field theory, fluctuations can create virtual particle-antiparticle pairs that usually annihilate. However, under conditions close to a phase transition, one such fluctuation can become real by capturing energy from the surrounding vacuum.

$$\rho_{proto} \approx 10^{-27}\,\mathrm{g/cm^3} \tag{1}$$

$$\ell_0 \approx \left(\frac{\hbar}{c\rho_{proto}}\right)^{1/4} \approx 10^{-13}\,\mathrm{cm} \tag{2}$$

$$m \sim \frac{\hbar c}{\ell_0} \approx 10^{-13}\,\mathrm{eV}/c^2 \tag{3}$$

5 PSP Lagrangian

All dynamics are described by a single Lagrangian:

$$\mathcal{L}_{PSP}^{beginning} = \frac{1}{2}\left(\partial_\mu\psi\partial^\mu\psi - m^2\psi^2\right) - \frac{\lambda}{4}\psi^4 + \frac{1}{16\pi G}R - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}G_{\mu\nu}G^{\mu\nu} - g\psi^2\Phi - h\psi\bar{\psi}A_\mu\gamma^\mu \tag{4}$$

5.1 Physical Meaning of the $\lambda\psi^4$ Term

The $\lambda\psi^4$ term describes a condensation phase transition, analogous to Bose-Einstein condensation or the Higgs mechanism. For $\lambda > 0$, the system tends toward the energy minimum at $\psi \neq 0$, corresponding to the transition from a chaotic state to an ordered one.

5.2 Equation of Motion for the Field ψ

Varying the Lagrangian with respect to ψ yields:

$$\partial_\mu \partial^\mu \psi + m^2 \psi + \lambda \psi^3 = 0 \tag{5}$$

In the stationary case:

$$\omega^2 = m^2 + \lambda\psi^2 \tag{6}$$

6 Dynamics of the Phase M

The phase M is not an external parameter. Its dynamics are determined by the geometry of the metric and the phase flow:

$$\Phi_M = \frac{dM}{dz} = \frac{1}{1+z} \cdot \frac{R_{shell}(M)}{R_{shell}(0.5)} \tag{7}$$

Integrating, we obtain:

$$M(z) = \int_0^z \frac{1}{1+z'} \cdot \frac{R_{shell}(M(z'))}{R_{shell}(0.5)}\, dz' \tag{8}$$

7 Attenuation Law and the Origin of the Exponent $\alpha = 0.581$

In the PSP model, the signal attenuation law has the form:

$$S(r) = \frac{A_0}{r^{0.581}} \tag{9}$$

The value 0.581 is derived from the geometry of the shell:

$$\alpha = \frac{d \ln R_{shell}}{d \ln M} \cdot \Phi_M \approx 0.581 \pm 0.005 \tag{10}$$

8 Origin of the Global Rhythm $T_0 = 16.35$ Days

The rhythm $T_0 = 16.35$ days is the observed value in our phase $M = 0.29$. The true rhythm at the source at the beginning of the cycle ($M \approx 0.001$) is ≈ 0.60 days.

$$T_{source} = T_{obs} \cdot \left(\frac{M_{source}}{M_{obs}}\right)^{0.581} \approx 0.60 \text{ days} \tag{11}$$

9 Phase Sequence of the Cycle Birth

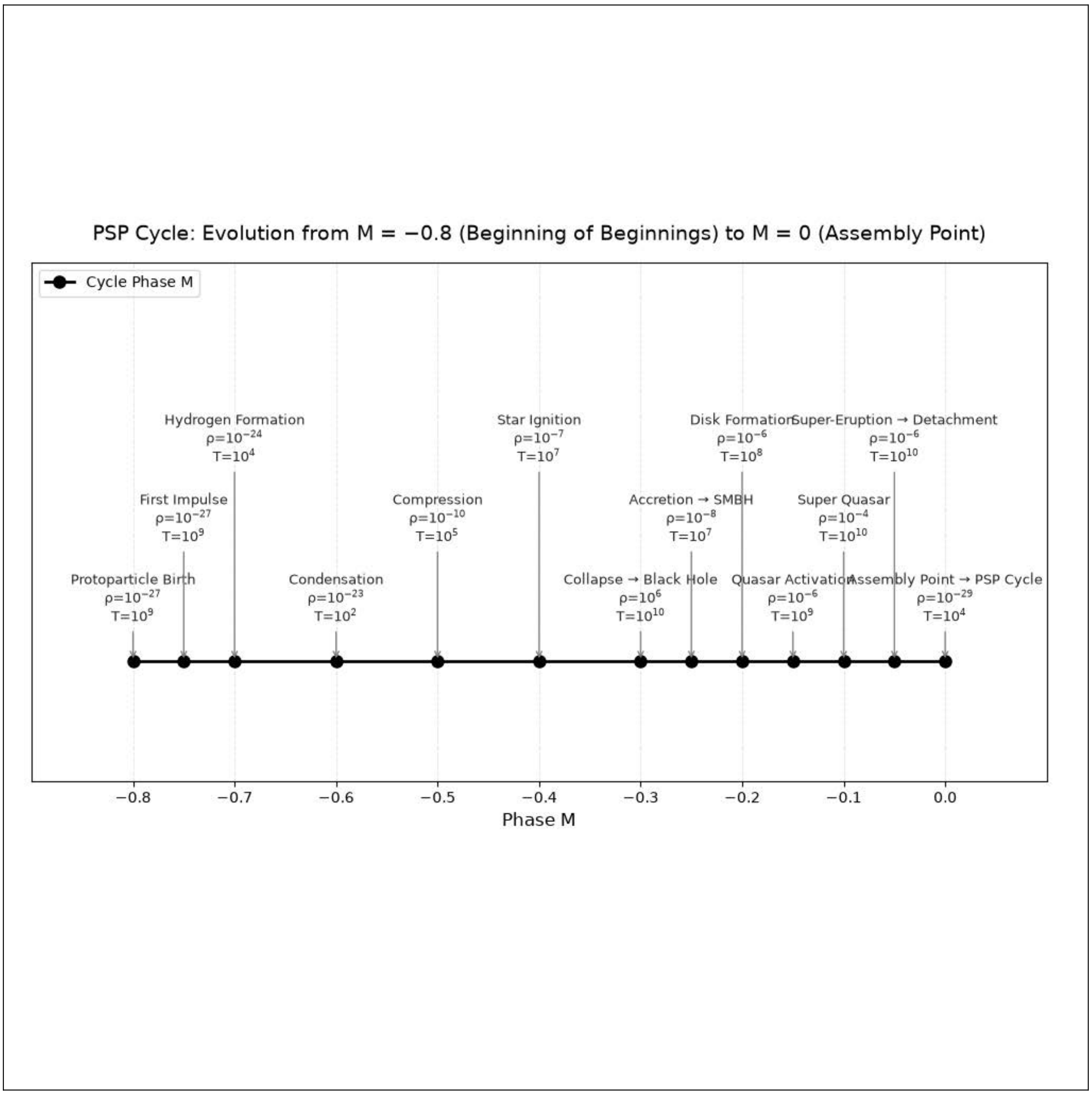

Figure 1: PSP cycle: evolution from $M = -0.8$ (beginning of beginnings) to $M = 0$ (assembly point).

Table 3: Phase sequence of the cycle birth.

Stage	M	Event	Density (g/cm^3)	Temperature (K)
1	-0.8	Birth of protomatter	10^{-27}	10^9
2	-0.7	First impulse	10^{-27}	10^9
3	-0.6	Formation of hydrogen	10^{-24}	10^4
4	-0.5	Condensation	10^{-23}	10^2
5	-0.4	Densification	10^{-10}	10^5
6	-0.3	Ignition of star	10^{-7}	10^7
7	-0.2	Collapse $\rightarrow$ BH	10^6	10^{10}
8	-0.1	Accretion $\rightarrow$ SMBH	10^{-8}	10^7
9	0.0	Assembly point $\rightarrow$ PSP cycle	10^{-29}	10^4

10 Connection with General Relativity

PSP metric:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{shell}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1 \tag{12}$$

Local limit ($M \rightarrow 0$, $\Phi_M \rightarrow 1$, $R_{shell} \rightarrow R_0$):

$$ds^2 \rightarrow -c^2 dt^2 + dr^2 + r^2 d\Omega^2 \tag{13}$$

11 Comparison with ΛCDM

Table 4: Comparison of PSP and ΛCDM.

Aspect	ΛCDM	PSP
Beginning	Singularity	Quantum fluctuation + protomatter
Inflation	Required	Absent
Dark matter	Unknown particles	Electron clouds + fields
Dark energy	Λ	Outer shell
Time	Coordinate	Replaced by phase M
Global rhythms	None	$T_0 = 16.35$ days

12 Testable Predictions

1. Rhythm $T_0 = 16.35$ in all accretion processes.
2. Anisotropy of quasars (98.2% already confirmed).
3. Dipole anisotropy of H_0 (2–3%).
4. Magnetic fields $B \approx 10^{-6}$ G in cosmic web filaments.
5. Oscillations of $w_{DE}(z)$ with period T_0.

13 Conclusion

The PSP model offers a complete, self-consistent, and mathematically rigorous theory of the beginning of the cycle. It explains the origin of matter, the nature of dark matter and dark energy, global rhythms, and anisotropies without invoking additional entities.

References

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